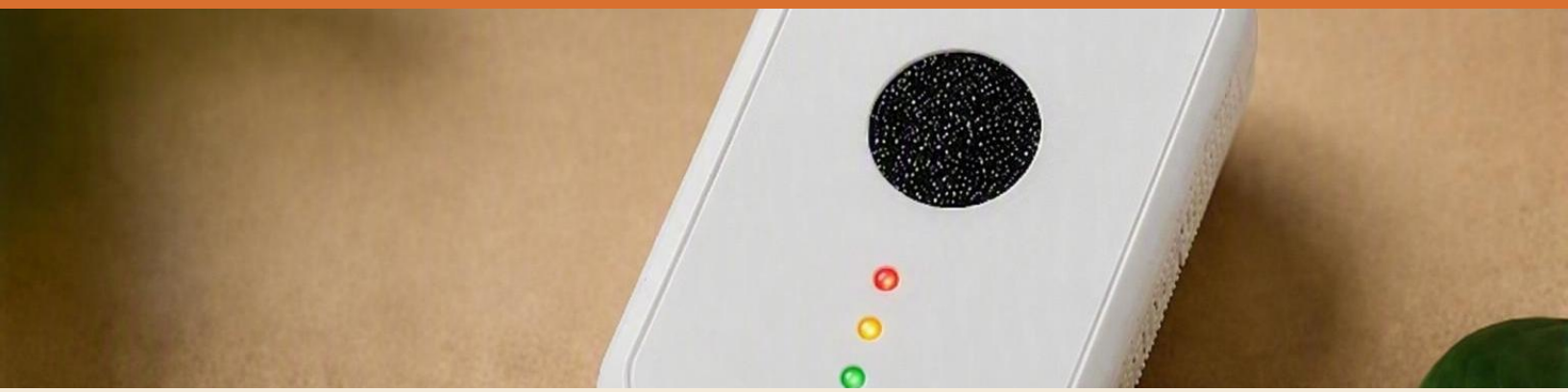
STUDENT INNOVATION · CLEAN AIR PROJECT

Portable CO_2 Detector

A pocket-sized air-quality sensor that tells you when the air you're breathing needs to change.



1 The Problem

Air quality impact on health, concentration, and comfort, requiring carbon dioxide (CO_2) monitor in educational, residential, and office places. Above 1,000 ppm, focus and memory drop, headaches set in, and poor ventilation lets airborne illness spread. Most people never see the number — professional monitors cost dollars and stay bolted to a wall.

2 The Solution

A palm-sized detector **CO_2 detector that could reliably monitor indoor air quality and provide immediate feedback.** An Co2 sensor samples the air every few seconds and a color-coded display shows green, yellow, or red with buzzer. Plug in to socket and act fast when needed.

3 Key Features

- LED lights with Co2 levels
- No battery required, USB-C charging
- Buzzer system, alert when red light
- Compact enclosed design
- No subscription, No maintenance

4 Community Impact

The detector turned an invisible problem into a daily habit: check the light, open a window. Houses, schools, and community centers get natural data to advocate for better ventilation — and students get a hands-on lesson in climate, health, and engineering, real problem solving.



MEET THE INVENTOR

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"You can't fix air you can't measure."

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Portfolio Website

Prototype figures are illustrative and based on the current pilot.